

Utkarsh Barethiya

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Authorization: Stamp 1G (Application in Progress)

PROFESSIONAL SUMMARY

Security Consultant and Penetration Tester with over 2.5 years' experience providing security assessments across Web Applications, APIs, Mobile Applications, Infrastructure and Active Directory for organisations in the enterprise and financial sectors. Extensive experience conducting Vulnerability Assessments, Red Team and Purple Team engagements, aligned to the OWASP Top 10 and MITRE ATT&CK framework. Holds a First Class Honours Master's degree in Cybersecurity, now applying offensive security expertise to Secure SDLC and Secure by Design practices through ForgeAI, an open-source AI Security Learning Platform.

EXPERIENCE

Anzen Technologies Pvt Ltd. – India, Navi Mumbai
Security Consultant / Penetration Tester

Aug 2022 - Dec 2024

Key Client Engagements: Leading Stock Exchange | Leading Banking Organisation

- Delivered Web Application, REST API, Network and Active Directory security assessments aligned to the OWASP Top 10 for major financial sector organisations, uncovering business logic, authorisation, authentication and input validation vulnerabilities, and producing technical assessment reports with risk based remediation guidance for engineering and security teams.
- Conducted internal and external network security assessments, identifying authentication and authorisation weaknesses, Kerberoasting opportunities, privilege escalation paths and lateral movement techniques, mapping attacker behaviour to the MITRE ATT&CK framework.
- Performed security assessments across 1,000+ internal and external IP addresses for a prominent banking institution, prioritising vulnerabilities using CVSS scoring, business impact analysis and risk based remediation strategies.
- Discovered, verified and documented 30+ high and medium risk vulnerabilities across 20+ enterprise applications and infrastructure assets, developing exploit proof of concepts and attack paths to demonstrate technical impact, justify security discussions with stakeholders and confirm the efficacy of security controls.
- Worked alongside SOC and Blue Team analysts on Purple Team exercises and cyber drills, analysing telemetry from Splunk, Forcepoint, Windows Event Logs, Cisco Firewalls, EDR, DLP and Palo Alto to validate alerts, investigate security incidents and improve detection and response capabilities.
- Embedded security across the Secure SDLC from threat modelling with developers and stakeholders during design, through security testing and remediation verification pre release, to working with cross functional engineering teams on secure coding recommendations and remediation prioritisation.
- Led security assessments to CERT-In prescribed national security testing standards, producing technical reports for CERT-In empanelled engagements.

Open-Source Project

ForgeAI – AI Security Learning Platform (GitHub)

- Design and development of an open source platform (Next.js, NestJS, PostgreSQL, Docker) to bind deliberately vulnerable AI apps with secure app versions to educate people on AI/LLM security practices.
- Implementing secured authentication (JWT, refresh tokens, RBAC), Secured API design, secure defense-in-depth using Secure by Design practices.
- Integrating GitHub actions, SAST, SCA, Dependency Scanning into the CI/CD security pipelines.

BUG BOUNTY & RESPONSIBLE DISCLOSURE:

- Identified and responsibly disclosed vulnerabilities in production systems of organizations including Dell, Swiggy, BigBasket & others and received multiple Hall of Fame recognitions for responsible disclosure contributions.

TECHNICAL SKILLS

- **Offensive Security:** Penetration Testing, Vulnerability Assessment (VAPT), Red Teaming, Adversary Simulation, Purple Teaming.
- **Application Security:** Web Application Security, API Security, OWASP Top 10, Authentication & Authorisation Testing, Business Logic Testing.
- **Network & AD Security:** Active Directory Exploitation, Privilege Escalation, Lateral Movement.
- **SIEM & Monitoring:** Splunk Enterprise, Windows Event Logging.
- **Frameworks:** MITRE ATT&CK, OWASP Top 10, CVSS and NIST CSF.
- **Security Practices:** Vulnerability Management, Security Risk Assessment, Secure Remediation Guidance, Security Reporting, Secure SDLC and Threat Modelling.

TOOLS & TECHNOLOGIES

Burp Suite Professional, Metasploit, Nmap, Nessus Professional, Tenable, Mimikatz, BloodHound, SharpHound, Wireshark, SQLMap and others.

EDUCATION

MSc in Cybersecurity (First Class Honours)
Dublin Business School

04/2025 - 06/2026
Dublin, Ireland

Project

Enterprise Active Directory Attack Simulation | MSc Dissertation

Evaluating Phishing Initiated Active Directory Compromise Paths in SMEs: A Simulation Based Red and Blue Team Risk Assessment Framework

- Designed and implemented an enterprise Active Directory lab using Windows Server, Windows clients and Kali Linux to simulate phishing initiated attacks in an SME environment. Conducted end to end attack simulations including Active Directory enumeration, privilege escalation, credential access, Kerberoasting and lateral movement using industry standard offensive security tools.

- Configured Splunk Enterprise with Universal Forwarders to centralised Windows Security Event Logs and monitor attacker activity across multiple systems.
- Developed SIEM detection queries for authentication events, privilege escalation, process creation and remote execution using Windows Event IDs 4624, 4672 and 4688.
- Validated detection effectiveness using Windows Event Logs, PowerShell logging and behavioural analysis to improve visibility into post compromise activity. Mapped attacker techniques to the MITRE ATT&CK framework to analyse attack progression and evaluate detection coverage across each stage of the attack lifecycle.
- Developed a Red Team and Blue Team risk assessment framework to evaluate attack paths, detection coverage, logging effectiveness and security controls within the simulated SME environment.

CERTIFICATIONS

- Certified Red Team Operator (CRTO) - Zero-Point Security | In Progress
- Certified Red Team Professional (CRTP) - Altered Security
- Junior Penetration Tester (eJPT) - eLearnSecurity
- Certified Ethical Hacker (CEH v11) - EC-Council